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UNIVERSITY KARACHI
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LAB 01

ADVANCE STATISTICS

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Course: Advance Statistics
Département: Data Science
Semester:4
Section: A

Lab 01: Mean Deviation

1. What is Mean Deviation?

Mean Deviation (MD) tells us how far, on average, each value in a dataset is fr
om the **mean**.It shows the **spread** or **dispersion** of data.

Formula:

MD= ∑|xi_x|/n

Where:

xi = each value

x¯ = mean (average) of the dataset

n = number of values

|xi_x¯| = absolute deviation from mean

Example Dataset (Marks of Students)

Student	Marks
1	45
2	78
3	62
4	55
5	90
6	72
7	50

Step-by-Step Calculation

Step 1: Find Mean

Xbar =45+78+62+55+90+72+507\7=64.57

Step 2:

Deviation means:

Xi − xbar

Where:

xi = each student's mark

xbar = mean (64.57)

It simply tells us how far each student's mark is from the mean.

If positive → the mark is **above average**.

If negative → the mark is **below average**.

3.Deviation for Each Student

Student	Marks (xi)	Deviation (xi−xbar)
1	45	45 − 64.57 = **−19.57**
2	78	78 − 64.57 = **+13.43**
3	62	62 − 64.57 = **−2.57**
4	55	55 − 64.57 = **−9.57**
5	90	90 − 64.57 = **+25.43**
6	72	72 − 64.57 = **+7.43**
7	50	50 − 64.57 = **−14.57**

4. Why Take Absolute Deviation?

If we add these deviations, positives and negatives will **cancel out**.

To avoid cancellation, we take **absolute values** (ignore minus signs).

| Student | Deviation (xi−xbar) | Absolute Deviation (|xi−xbar|)

|---|---|---|

1	−19.57	19.57
2	+13.43	13.43
3	−2.57	2.57
4	−9.57	9.57
5	+25.43	25.43
6	+7.43	7.43
7	−14.57	14.57

5.Mean Deviation

Now take the **average** of the absolute deviations:

MD=19.57+13.43+2.57+9.57+25.43+7.43+14.577/7=13.51

Mean Deviation = 13.51

Mean Deviation in Excel (Step by Step)

Step 1: Enter Dataset

In **Excel**, enter the marks:

A (Student)	B (Marks)
1	45
2	78
3	62
4	55
5	90
6	72
7	50

Step 2: Find the Mean

In **cell B9**, type:

=AVERAGE(B2:B8)

This will give **64.57** (mean of marks).

Step 3: Calculate Deviation (xi – Mean)

In **C2**, type:

=B2 - B9

Press Enter.

Drag the formula down till **C8**.

This gives each student's deviation (positive or negative).

Step 4: Absolute Deviation

In **D2**, type:

=ABS(C2)

Press Enter

Drag down till **D8**.

Now all deviations are positive.

Step 5: Mean Deviation

In **D9**, type:

=AVERAGE(D2:D8)

This gives **13.51** (the Mean Deviation).

Final Excel Table

Student	Marks (B)	Deviation (C = B − Mean)	Absolute Deviation (D = ABS(C))
1	45	-19.57	19.57
2	78	13.43	13.43
3	62	-2.57	2.57
4	55	-9.57	9.57
5	90	25.43	25.43
6	72	7.43	7.43
7	50	-14.57	14.57
Mean	**64.57**		**13.51 (Mean Deviation)**

LAB TASK

To understand and calculate **Mean Deviation** using Microsoft Excel.

Task Instructions:

Enter Data in Excel
Enter the following dataset of student marks in Excel:

Student	Marks
1	12
2	18
3	20
4	25
5	30
6	22
7	15

Find the Mean of Marks

Use the formula:

=AVERAGE(B2:B8)

Calculate Deviation (Marks – Mean)

In column C, subtract the mean from each mark.

Formula (for first student):

=B2 - B9

Find Absolute Deviation

In column D, take the absolute value of each deviation.

Formula:

=ABS(C2)

Calculate Mean Deviation

In the last cell, take the average of all absolute deviations.

Formula:

=AVERAGE(D2:D8)

Draw a Chart

Insert a **Column Chart** for the absolute deviations to show how far each stude nt's marks are from the mean.